$$y = mx + c \qquad \ldots (2)$$

If we substitute $mx + c$ for y in (1), we get

$$x^2 + (mx + c)^2 - r^2 = 0$$

which is same as the quadratic equation

$$(1 + m^2)x^2 + 2mcx + (c^2 - r^2) = 0\,. \qquad \ldots (3)$$

This equation cannot have more than two solutions, and hence a line and a circle cannot intersect at more than two points. ■

It is interesting to note that a substitution makes the problem of solving a system of two equations in two variables into a problem of solving a quadratic equation.

Further we note that as the coefficients of the reduced quadratic polynomial are real, either both roots are real or both imaginary. If both roots are imaginary numbers, we conclude that the circle and the straight line do not intersect. In the case of real roots, either they are distinct or multiple roots of the polynomial. If they are distinct, substituting in (2), we get two values for y and hence two points of intersection. If we have equal roots, we say the straight line touches the circle as a tangent. As the polynomial (3) cannot have only one simple real root, a line cannot cut a circle at only one point.

Note

A technique similar to the one used in example 3.14 may be adopted to prove

- two circles cannot intersect at more than two points.
- a circle and an ellipse cannot intersect at more than four points.

EXERCISE 3.2

1. If k is real, discuss the nature of the roots of the polynomial equation $2x^2 + kx + k = 0$, in terms of k.
2. Find a polynomial equation of minimum degree with rational coefficients, having $2 + \sqrt{3}i$ as a root.
3. Find a polynomial equation of minimum degree with rational coefficients, having $2i + 3$ as a root.
4. Find a polynomial equation of minimum degree with rational coefficients, having $\sqrt{5} - \sqrt{3}$ as a root.
5. Prove that a straight line and parabola cannot intersect at more than two points.

3.6 Roots of Higher Degree Polynomial Equations

We know that the equation $P(x) = 0$ is called a polynomial equation. The root or zero of a polynomial equation and the solution of the corresponding polynomial equation are the same. So we use both the terminologies.

We know that it is easy to verify whether a number is a root of a polynomial equation or not, just by substitution. But when finding the roots, the problem is simple if the equation is quadratic and it is in general not so easy for a polynomial equation of higher degree.

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